CS 479

Programming Assignment 1, Bayesian Decision Theory

Jackson Loughmiller

3/16/2026

I declare that all material in this assignment is my own work except where there is clear acknowledgment or reference to the work of others. I understand that both my report and code may be subjected to plagiarism detection software, and fully accept all consequences if found responsible for plagiarism, as explained in the syllabus, and described in UNR's Academic Standards Policy: UAM 6,502.

Theory

Experiment 1: Bayesian Decision Theory Case I

Conditional Probability

In Bayesian Decision Theory, a certain class can be thought of as a state of nature, usually denoted with **w**. With this notation, a system with 2 possible states of nature would contain $\mathbf{w_1}$ and $\mathbf{w_2}$. The prior probability, **P(w)**, is a representation of the prior knowledge we have of the probability of each class occurring, before we attempt to classify anything. The features of each class, **x**, and their pdf, **p(x)**, denote the pdf of each feature that is present in the data with no relation to which class the features originated from.

The conditional pdf, $\mathbf{p(x/w_j)}$, represents the *likelihood* that a certain feature is present given the state of nature is $\mathbf{w_j}$. This value is distinct from the conditional probability, $\mathbf{P(w_j/x)}$, which represents the *probability* that a certain instance belongs to class $\mathbf{w_j}$ given the feature **x**.

Using only the information stated so far, one can create a simple classifier using the following equation, eq. 1.1:

$$P(w_j/x) = \frac{p(x/w_j)P(w_j)}{p(x)} = \frac{likelihood \times prior}{evidence} \qquad \text{Eq. 1.1}$$

Effectively, this classifier will choose $\mathbf{w_1}$ if $\mathbf{P(w_1/x) > P(w_2/x)}$, otherwise it will choose $\mathbf{w_2}$. This is known as the decision rule using posterior probabilities. This classifier is fundamentally limited, as it does not take into account other possible costs associated with making certain classification decisions. Therefore, a more robust theory is needed to take these costs into consideration.

Conditional Risk

A more general theory of classification is required that takes into account that there may be multiple features or categories, that allows actions other than classifications (such as a reject action if the model is uncertain of the correct class), that associates costs with each action, and that works by way of minimizing the overall cost, rather than maximizing the conditional probability.

This theory considers a set of **c** categories $\mathbf{w_1, w_2, \ldots, w_c}$, a finite set of **l** actions $\boldsymbol{\alpha_1, \alpha_2, \ldots, \alpha_l}$, a loss function $\boldsymbol{\lambda(\alpha_i/w_j)=\lambda_{ij}}$ (representing the cost associated with taking an action $\boldsymbol{\alpha_i}$ given the correct class is $\mathbf{w_j}$), and the conditional risk $\mathbf{R(\alpha_i/x)}$ (the expected loss of taking action $\boldsymbol{\alpha_i}$ given

x). This theory works by way of minimizing **$R(\alpha_i/x)$** rather than maximizing **$P(w_i/x)$.** This allows us to configure the classifier to make it less likely that it will make detrimental classification decisions. This is important especially if the classifier is used for use cases where security is a great concern.

The conditional risk is given by the following equation, Eq. 1.2:

$$R(\alpha_i/x) = \sum_{j=1}^{c} \lambda(\alpha_i/w_j)P(w_j/x),\ i = 1, 2,..., l \qquad \text{Eq. 1.2}$$

The decision rule using conditional risk is to decide α_i if **$R(\alpha_i/x) < R(\alpha_j/x)$ for every $j \neq i$.** This effectively minimizes the conditional risk, as opposed to maximizing the conditional probability. A special case using the conditional risk for classification occurs when the same cost is assigned to all classification errors, this is called the Zero-One Loss Function. In this case, the decision rule becomes equivalent to the Bayes Decision Rule using posterior probabilities, mentioned before.

Discriminant Functions

Classifiers can be represented with a set of discriminant functions, using one such function per class. Assuming a zero-one loss function, the discriminant function can be represented by the following equation, Eq. 1.3:

$$g_i(x) = P(w_i/x) = p(x/w_i)P(w_i) \qquad \text{Eq. 1.3}$$

Assuming a general loss function, the discriminant can be represented by Eq. 1.4:

$$g_i(x) = -R(\alpha_i/x) \qquad \text{Eq. 1.4}$$

The same results can be yielded from Eq. 1.3 by replacing **$g_i(x)$** with **$f(g_i(x))$**, where **f()** is a monotonically increasing function. Generally, the natural logarithmic function is used to simplify the calculations required, as shown in Eq. 1.5:

$$g_i(x) = ln(p(x/w_i)) + ln(P(w_i)) \qquad \text{Eq. 1.5}$$

These discriminant functions divide the feature space into different decision regions by creating decision boundaries. There are 3 possible cases when creating discriminant functions, each of which allows for differing amounts of simplification of the equation. The full equation for the discriminant assuming that **$p(x/w_i) \sim N(\mu_i, \Sigma_i)$** is expressed by Eq. 1.6:

$$g_i(x) = -\frac{1}{2}(x - \mu_i)^t \Sigma_i^{-1}(x - \mu_i) - \frac{d}{2} ln 2\pi - \frac{1}{2} ln\left|\Sigma_i\right| + ln(P(w_i)) \qquad \text{Eq. 1.6}$$

Discriminant Case I

Case I is a case in which each class **w_i** is modeled by the same covariance matrix, that is diagonal with equal values. The features in this case are uncorrelated, and have the same variance, and the clusters have a spherical shape and are the same size. The discriminant in this case can be simplified to Eq. 1.7:

$$g_i(x) = -\frac{||x-\mu_i||^2}{2\sigma^2} + ln(P(w_i)) \qquad \text{Eq. 1.7}$$

Expanding the above equation and disregarding the constant **$x^t x$,** we get the linear discriminant shown in Eq. 1.8:

$$g_i(x) = w_i^t x + w_{i0} \qquad \text{Eq. 1.8}$$

The decision boundary for this and any case can be found by setting **$g_i(x)=g_j(x)$.** The decision boundary for this case is determined by hyperplanes, by extending a plan from the linear function orthogonal to the line connecting the two means.

Discriminant Case II

Case II is a case in which the covariance matrices are all the same, however they are not necessarily diagonal. The equation for the discriminant in this case can only be simplified to Eq. 1.9:

$$g_i(x) = -\frac{1}{2}(x - \mu_i)^t \Sigma_i^{-1}(x - \mu_i) + ln(P(w_i)) \qquad \text{Eq. 1.9}$$

This equation can be shown to also be linear discriminant, with the same equation seen in Eq. 1.8. Similar to case I, its decision boundary is also determined by hyperplanes, however the hyperplane is not orthogonal to the line connecting the two means.

Error Bounds

Error bounds are important for estimating the theoretical error that a model could have. Since the exact error for a model is extremely difficult to compute, we use an upper bound for the error instead that indicates a value for which the error will never be higher. The equation for the error bound of a **p(x/w$_i$)** Gaussian distribution is given in Eq. 1.10, below:

$$P(error) \leq P^{\beta}(w_1)P^{1-\beta}(w_2)e^{-k(\beta)} \qquad \text{Eq. 1.10}$$

There are two main ways of evaluating this expression. The first is the Chernoff error bound, which is computed by minimizing $e^{-k(\beta)}$. The second is the Bhattacharyya error bound which is computed by selecting β=0.5. Typically the Chernoff bound is tighter than the Bhattacharyya bound, meaning it is likely closer to the actual error, while the Bhattacharyya is likely higher than the actual error.

Experiment 2

Discriminant Case III

Case III is a case in which the covariance matrices are arbitrary, meaning each state of nature has its own covariance matrix. The clusters have different shapes and sizes. In this case, the discriminant is quadratic, not linear, and is given by the following equation, Eq. 1.11:

$$g_i(x) = x^t W_i x + w_i^t x + w_{i0} \qquad \text{Eq. 1.11}$$

where

$$W_i = -\frac{1}{2}\Sigma_i^{-1},\ w_i = \Sigma_i^{-1}\mu_i,\ and\ w_{i0} = -\frac{1}{2}\mu_i^t\Sigma_i^{-1}\mu_i - \frac{1}{2}ln\left|\Sigma_i\right| + ln(P(w_i)$$

The decision boundary for Case III is determined by hyperquadrics, as opposed to hyperplanes.

Experiments 3 & 4

Euclidean Distance Classifier

If **P(w$_i$)** are all equal, the Eq. 1.7 can be further simplified to Eq. 1.12:

$$g_i(x) = -\ ||x - \mu_i||^2 \qquad \text{Eq. 1.12}$$

This is known as the Euclidean Distance Classifier. This classifier is optimal only if the prior probabilities are equal among all classes.

Results and Discussion

Experiment 1

Part 1.a

For this experiment, the case I discriminant is the optimum function to use because the covariance matrices are both equal, and diagonal. The prior probability for **w$_1$** and **w$_2$** are shown below in Eq. 2.1:

$$P(w_1) = \frac{60{,}000}{60{,}000+140{,}000} = 0.3,\ P(w_2) = \frac{140{,}000}{60{,}000+140{,}000} = 0.7 \qquad \text{Eq. 2.1}$$

Part 1.b

Below, in Figure 2.1, the Bayes decision boundary is plotted along with the samples from dataset A.

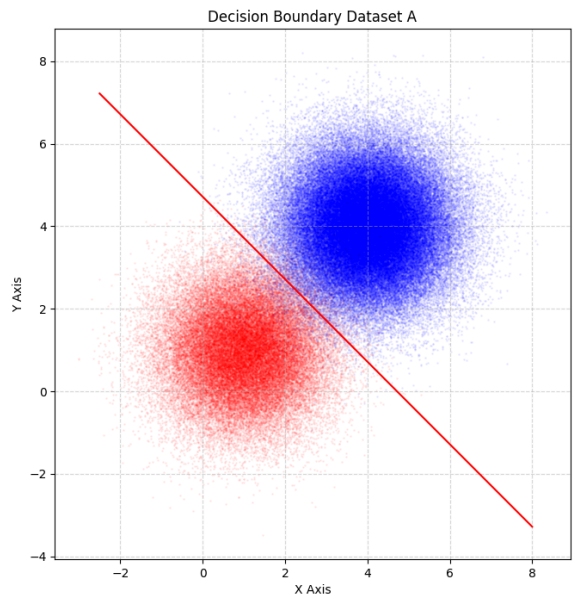

Figure 2.1: Plot of dataset A, w_1 (red) and w_2 (blue), and the decision boundary.

Part 1.c

After classifying the samples, the following misclassification rates were calculated:

$\mathbf{w_1}$: 2.75%

$\mathbf{w_2}$: 1.02%

Total: 1.54%

These rates show the model misclassified $\mathbf{w_1}$ more often than $\mathbf{w_2}$, indicating there is a larger overlap of $\mathbf{w_2}$'s points on $\mathbf{w_1}$'s side of the decision boundary. This is likely due to there being a significantly larger number of data points for $\mathbf{w_2}$.

Part 1.d

The Bhattacharyya error bound was calculated using Eq. 1.10, giving a value of 4.83%. This value makes sense given the misclassification rates, as it is higher than the total misclassification rate. The bound is more than double the actual rate, due to using the Bhattacharyya bound. The Chernoff bound would have given an answer closer to the true error rate.

Experiment 2

Part 2.a

For this experiment, the case III discriminant is the optimum function to use because the covariance matrices are not equal. The prior probability for $\mathbf{w_1}$ and $\mathbf{w_2}$ are shown below in Eq. 2.2:

$$P(w_1) = \frac{60{,}000}{60{,}000+140{,}000} = 0.3,\ P(w_2) = \frac{140{,}000}{60{,}000+140{,}000} = 0.7 \qquad \text{Eq. 2.2}$$

Part 2.b

Below, in Figure 2.2, the Bayes decision boundary is plotted along with the samples from dataset B.

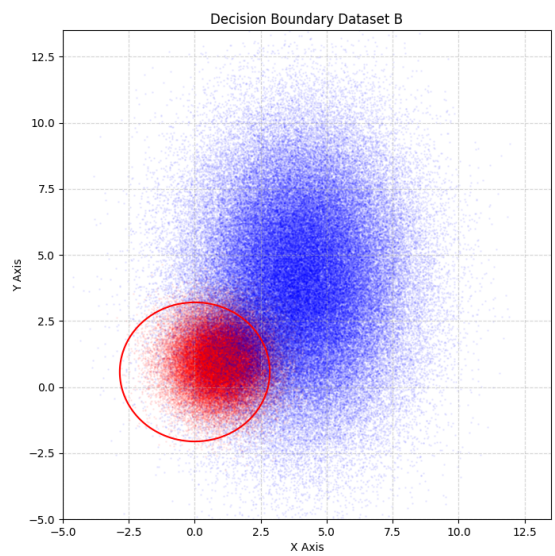

Figure 2.2: Plot of dataset B, w_1 (red) and w_2 (blue), and the decision boundary.

Part 2.c

After classifying the samples, the following misclassification rates were calculated:

$\mathbf{w_1}$: 15.77%

$\mathbf{w_2}$: 8.42%

Total: 10.63%

These rates show the model misclassified $\mathbf{w_1}$ more often than $\mathbf{w_2}$, this is likely due to $\mathbf{w_1}$'s distribution being entirely inside of $\mathbf{w_2}$'s.

Part 2.d

The Bhattacharyya error bound was calculated using Eq. 1.10, giving a value of 16.41%. This value makes sense given the misclassification rates, as it is higher than the total misclassification rate. The bound is significantly larger than the actual rate, due to using the Bhattacharyya bound. The Chernoff bound would have given an answer closer to the true error rate.

Experiment 3

The results for using the Euclidean Distance Classifier on dataset A are similar to the Bayesian classifier, however the decision boundary is moved closer to $\mathbf{w_2}$, meaning the classifier is more likely to choose $\mathbf{w_1}$, as shown in Figure 2.3 below.

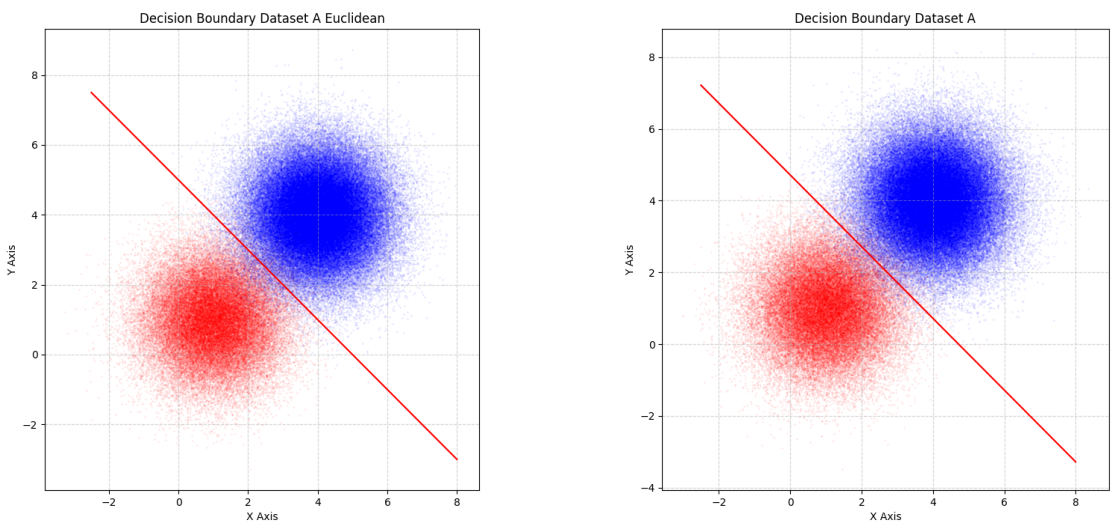

Figure 2.3: Decision boundary for dataset A using Euclidean Distance Classifier (left) and Bayesian Classifier (right).

The misclassification rates for the Euclidean Distance Classifier on dataset A are shown below:

Euclidean	**Bayesian**
$\mathbf{w_1}$: 2.75%	$\mathbf{w_1}$: 1.73%
$\mathbf{w_2}$: 1.02%	$\mathbf{w_2}$: 1.7%
Total: 1.54%	**Total**: 1.71%

As expected, the error became lower for $\mathbf{w_1}$, higher for $\mathbf{w_2}$, and higher for the total error.

Experiment 4

The results for using the Euclidean Distance Classifier on dataset B produced significantly worse results than the Bayesian classifier. This is not surprising, as the Bayesian classifier made use of a hyperquadric decision boundary, not a linear one. The decision boundary using Euclidean Distance is shown below in Figure 2.4:

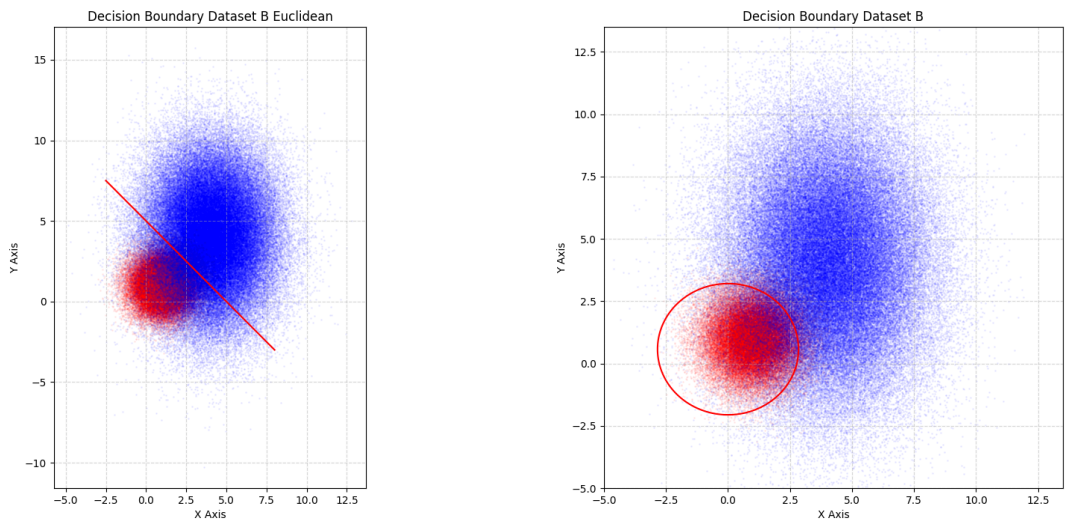

Figure 2.4: Decision boundary for dataset B using Euclidean Distance Classifier (left) and Bayesian Classifier (right).

The misclassification rates for the Euclidean Distance Classifier on dataset B is shown below:

Euclidean	**Bayesian**
$\mathbf{w_1}$: 1.67%	$\mathbf{w_1}$: 15.77%
$\mathbf{w_2}$: 19.5%	$\mathbf{w_2}$: 8.42%
Total: 14.153%	**Total**: 10.63%

The misclassification rate went down significantly for $\mathbf{w_1}$, since almost all of its data points are on its side of the decision boundary. However, the misclassification rate for $\mathbf{w_2}$ more than doubled, and the total misclassification rate went up by about 4%.